

Ausgewählte SQL-Befehle (nicht vollständig)

Kategorie	Syntax / Beispiel	Beschreibung
Querying Data	SELECT c1, c2 FROM t;	Spalten abfragen
Querying Data	SELECT * FROM t;	Alle Spalten/Zeilen abfragen
Querying Data	SELECT c1, c2 FROM t WHERE condition;	Zeilen filtern
Querying Data	SELECT DISTINCT c1 FROM t WHERE condition;	Nur eindeutige Werte
Querying Data	SELECT c1, aggregate(c2) FROM t GROUP BY c1;	Gruppieren
Querying Data	SELECT c1, aggregate(c2) FROM t GROUP BY c1 HAVING condition;	Gruppen filtern
Querying Data	SELECT c1, c2 FROM t ORDER BY c1 ASC [DESC];	Sortieren
Querying Data	SELECT c1, c2 FROM t ORDER BY c1 LIMIT n OFFSET offset;	Pagination

Joins & Multiple Tables	SELECT c1, c2 FROM t1 INNER JOIN t2 ON condition;	Inner Join
Joins & Multiple Tables	SELECT c1, c2 FROM t1 LEFT JOIN t2 ON condition;	Left Join
Joins & Multiple Tables	SELECT c1, c2 FROM t1 RIGHT JOIN t2 ON condition;	Right Join
Joins & Multiple Tables	SELECT c1, c2 FROM t1 FULL OUTER JOIN t2 ON condition;	Full Outer Join

Set-Operatoren	... UNION [ALL] ...	Ergebnismengen vereinen
Set-Operatoren	... INTERSECT ...	Schnittmenge
Set-Operatoren	... EXCEPT ...	Differenz

Where-Operatoren	c1 [NOT] LIKE pattern	Mustervergleich
Where-Operatoren	c1 [NOT] IN (...)	Listenprüfung
Where-Operatoren	c1 BETWEEN low AND high	Bereich
Where-Operatoren	c1 IS [NOT] NULL	Null-Wertprüfung

Managing Tables	CREATE TABLE t (...);	Tabelle erstellen
Managing Tables	ALTER TABLE t ADD column;	Spalte hinzufügen
Managing Tables	ALTER TABLE t DROP COLUMN c;	Spalte löschen
Managing Tables	ALTER TABLE t ALTER COLUMN c TYPE VARCHAR(100);	Spalte ändern
Managing Tables	ALTER TABLE t ADD CONSTRAINT chk_value CHECK (c > 0);	Constraint hinzufügen (hier Check)
Managing Tables	ALTER TABLE t1 RENAME TO t2;	Tabelle umbenennen
Managing Tables	ALTER TABLE t1 RENAME c1 TO c2;	Spalte umbenennen
Managing Tables	DROP TABLE t CASCADE;	Tabelle löschen

Modifying Data	INSERT INTO t (...) VALUES (...);	Einfügen
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Modifying Data	INSERT INTO t VALUES (...), (...);	Mehrere Zeilen
Modifying Data	INSERT INTO t1 SELECT ... FROM t2;	Aus anderer Tabelle einfügen
Modifying Data	UPDATE t SET ... WHERE condition;	Zeilen aktualisieren
Modifying Data	DELETE FROM t WHERE ...;	Zeilen löschen

Constraints	PRIMARY KEY (c1,c2)	Primärschlüssel
Constraints	FOREIGN KEY (c2) REFERENCES t2(c2)	Fremdschlüssel
Constraints	UNIQUE(c2,c3)	Eindeutigkeit
Constraints	CHECK(c1 > 0 AND c1 >= c2)	Gültige Werte
Constraints	c2 VARCHAR NOT NULL	Pflichtfeld

Indexes	CREATE INDEX idx_name ON t(c1,c2);	Index
Indexes	CREATE UNIQUE INDEX idx_name ON t(c3,c4);	Einzigartiger Index
Indexes	DROP INDEX idx_name;	Index löschen

Views	CREATE VIEW v AS SELECT ...;	View erstellen
Views	CREATE VIEW v AS ... WITH CHECK OPTION;	View mit Prüfung
Views	DROP VIEW view_name;	View löschen

Triggers	CREATE TRIGGER ... BEFORE INSERT ON t ...;	Trigger erzeugen
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Triggers	DROP TRIGGER trigger_name;	Trigger löschen
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Aggregate Functions	AVG(), COUNT(), SUM(), MAX(), MIN()	Aggregatfunktionen
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CREATE FUNCTION

```
CREATE [ OR REPLACE ] FUNCTION
    name ( [ [ argmode ] [ argname ] argtype [ { DEFAULT | = } default_expr ] [, ...] ] )
    [ RETURNS rettype
      | RETURNS TABLE ( column_name column_type [, ...] ) ]
    { LANGUAGE lang_name
      | TRANSFORM { FOR TYPE type_name } [, ... ]
      | WINDOW
      | { IMMUTABLE | STABLE | VOLATILE }
      | [ NOT ] LEAKPROOF
      | { CALLED ON NULL INPUT | RETURNS NULL ON NULL INPUT | STRICT }
      | { [ EXTERNAL ] SECURITY INVOKER | [ EXTERNAL ] SECURITY DEFINER }
      | PARALLEL { UNSAFE | RESTRICTED | SAFE }
      | COST execution_cost
      | ROWS result_rows
      | SUPPORT support_function
      | SET configuration_parameter { TO value | = value | FROM CURRENT }
      | AS 'definition'
      | AS 'obj_file', 'link_symbol'
      | sql_body
    } ...
```

CREATE PROCEDURE

```
CREATE [ OR REPLACE ] PROCEDURE
    name ( [ [ argmode ] [ argname ] argtype [ { DEFAULT | = } default_expr ] [, ...] ] )
    { LANGUAGE lang_name
      | TRANSFORM { FOR TYPE type_name } [, ... ]
      | [ EXTERNAL ] SECURITY INVOKER | [ EXTERNAL ] SECURITY DEFINER
      | SET configuration_parameter { TO value | = value | FROM CURRENT }
      | AS 'definition'
      | AS 'obj_file', 'link_symbol'
      | sql_body
    } ...
```

CREATE TRIGGER

```
CREATE [ OR REPLACE ] [ CONSTRAINT ] TRIGGER name { BEFORE | AFTER | INSTEAD OF } { event [ OR ... ] }  
  ON table_name  
  [ FROM referenced_table_name ]  
  [ NOT DEFERRABLE | [ DEFERRABLE ] [ INITIALLY IMMEDIATE | INITIALLY DEFERRED ] ]  
  [ REFERENCING { { OLD | NEW } TABLE [ AS ] transition_relation_name } [ ... ] ]  
  [ FOR [ EACH ] { ROW | STATEMENT } ]  
  [ WHEN ( condition ) ]  
  EXECUTE { FUNCTION | PROCEDURE } function_name ( arguments )
```

where *event* can be one of:

```
INSERT  
UPDATE [ OF column_name [, ... ] ]  
DELETE  
TRUNCATE
```